Signals & Systems:

Continuous Time Fourier Series (CTFS):

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T}$$

$$S_T = \left\{ x : \mathbb{R} \to \mathbb{C} \;\middle|\; x(t+T) = x(t)\ \forall t,\ \frac{1}{T}\int_{\langle T \rangle} |x(t)|^2 dt < \infty \right\}$$

To prove: 1) S_T is a V.S.

2) $\{e^{jk\omega_0 t}\}_{k=-\infty}^{\infty}$ is a basis of S_T

If x & $y \in S_T$

i) $x+y \in S_T$ & $\alpha x \in S_T$

Sum of 2 periodic signals.

$$\frac{1}{T}\int_{\langle T \rangle} |(x+y)(t)|^2 dt < \frac{1}{T}\int_{\langle T \rangle} (|x(t)| + |y(t)|)^2 dt = \frac{1}{T}\int \underbrace{\overbrace{|x(t)|^2}^{<\infty} + \overbrace{|y(t)|^2}^{<\infty} + 2|x(t)||y(t)|}_{<\infty} dt$$

$< \infty$

$=$

αx is periodic with T

$$\frac{1}{T}\int_{\langle T \rangle} |\alpha x(t)|^2 dt = |\alpha|^2 \underbrace{\int_{\langle T \rangle} |x(t)|^2 dt}_{P_x} < \infty$$

$\therefore S_T$ is a V.S.

$\{e^{jk\omega_0 t}\}_{k=-\infty}^{\infty}$ is a basis of S_T.

To prove: It is a linearly indep. set in S_T.

$$x_k(t) \triangleq e^{jk\omega_0 t}$$

$$\langle x_k, x_l \rangle = \begin{cases} 0 & , k \neq l \\ 1 & , k = l \end{cases} \qquad \langle x, y \rangle \triangleq \frac{1}{T}\int_{\langle T \rangle} x(t)\, y^*(t)\, dt$$

$$\langle x_k, x_l \rangle = \frac{1}{T}\int_{\langle T \rangle} x_k(t)\, x_l^*(t)\, dt = \frac{1}{T}\int_{\langle T \rangle} e^{jk\omega_0 t}\, e^{-jl\omega_0 t}\, dt$$

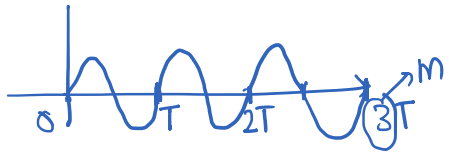
$$= \frac{1}{T}\int_{\langle T \rangle} e^{j\overset{m}{(k-l)}\omega_0 t}\, dt$$

$$= \begin{cases} 1 & , k = l \\ 0 & , k \neq l \end{cases}$$

$$\frac{1}{T}\int_{\langle T \rangle} e^{jm\omega_0 t}\, dt \quad , \omega_0 = \frac{2\pi}{T}$$

$$\underbrace{\frac{1}{T}\int_{\langle T \rangle} \cos(\overset{T/m}{m\omega_0 t}) + j\sin(\overset{T/m}{m\omega_0 t})\, dt}_{0+0\,=0}$$

$\therefore \{x_k\}_{k=-\infty}^{\infty}$ is an o.n. set in S_T.



$e^{jk\omega_0 t}$ is periodic with T & Power $= 1$ $\Big\} \in S_T$

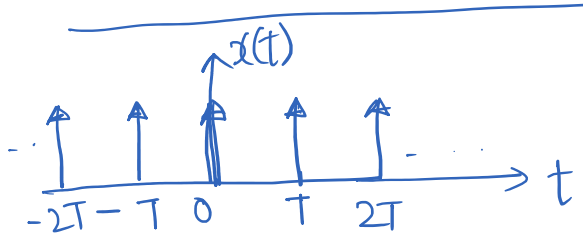
$\{e^{jk\omega_0 t}\}_{k=-\infty}^{\infty}$ forms a basis for S_T.

CONSET (Complete Orthonormal Set) $\}$ Basis of S_T

$$x(t) \in S_T \Rightarrow x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longrightarrow \text{CTFS (Synthesis Eq}^n\text{)}$$

As $k \uparrow \longrightarrow$ Freq. of $e^{jk\omega_0 t} \uparrow$

$a_k \longrightarrow$ Amount of freq. $k\omega_0$ in $x(t)$ $\Big\}$ Freq. Decompos^n of $x(t)$



$$x(t) = \sum a_k e^{jk\omega_0 t}$$

$$x = \sum_{k=-\infty}^{\infty} a_k x_k \quad ,$$

$$\langle x, x_p \rangle = \left\langle \sum_{k=-\infty}^{\infty} a_k x_k, x_p \right\rangle = a_p \langle x_p, x_p \rangle = a_p$$

$$a_p = \langle x, x_p \rangle \quad \therefore a_k = \langle x, x_k \rangle = \frac{1}{T}\int_{\langle T \rangle} x(t)\, e^{-jk\omega_0 t}\, dt$$

FS Coeff^n Eq^n

$T\langle t\rangle$

F.S. Coeff. F.S. Analysis Eq^n